SmellPredict: Dual-Engine Model Upgrade & Forensic Evaluation Report

Comparative Audit: Legacy Model (v1 Isotonic Single-Engine) vs. Upgraded System (v2 Dual-Engine Platt Sigmoidal)
Dataset Scope: 23,170 Enriched Commit Snapshots · 46 In-Pool LOPO Codebases · 4 Zero-Shot Reserved OOD Repositories (1,680 Snapshots)

1. Executive Summary & Forensic Audit Motivation

Following an exhaustive forensic audit across all 48 benchmark repositories and 23,170 snapshots, three fundamental failure modes were identified in the legacy v1 architecture that necessitated a complete dual-engine restructuring:

CORE MOTIVATIONS FOR ARCHITECTURAL UPGRADE

1. Isotonic Tail Overfitting: Legacy v1 employed non-parametric isotonic regression over small 5-fold calibration sets. This caused severe tail compression, producing uncalibrated probability collapse (0.000 or 1.000) on extreme inputs.
2. Live Editor Cold-Start Failure: v1 relied on a single 73-feature model requiring historical Git churn. In IDE buffers lacking Git history, heuristic zero-filling degraded predictive rank order and created inconsistent user priors.
3. Multi-Language Scope Leakage: Non-Python files (such as base64 images or polyglot source) inadvertently triggered Python AST metric parsers, yielding false high-risk defect probability scores on non-code assets.
4. Prevalence Inflation Clarification: Raw PR-AUC was confirmed to have a 0.887 correlation with repository defect prevalence, demonstrating that genuine model predictive power must be evaluated via **Lift (PR-AUC − Prevalence)**.

2. Architectural Paradigm Shift: v1 vs. v2

SmellPredict v2 splits inference into a decoupled **Dual-Engine Architecture** designed for specific deployment contexts:

Dimension	Legacy Model (v1)	Engine A (v2 Static AST)	Engine B (v2 Full Enterprise)
Target Environment	All contexts (Single model)	IDE Live Editor / Cold-Start	CI/CD Pipeline / PR Bot / Gating
Git Churn Dependency	Mandatory (Fails on untracked)	**Zero (Pure Static AST)**	Mandatory (Full Churn + Co-Change)
Feature Space	73 mixed features	**34 AST & Complexity Features**	**73 Full Multidimensional Features**
Calibration Method	Isotonic Regression (Overfits tails)	**Platt Sigmoid Scaling**	**Platt Sigmoid Scaling**
Regularization	Default L2 (reg_lambda=0)	reg_lambda=2.0, reg_alpha=0.5	reg_lambda=2.0, reg_alpha=0.5
Monotonic Constraints	None (Spurious inversions)	+1 Complexity, +1 Smells, -1 MI	+1 Complexity, +1 Smells, -1 MI
Quantile Normalization	Dynamic batch ranking only	**101-Point Empirical CDF Tables**	**101-Point Empirical CDF Tables**
Non-Python Isolation	Unchecked fallback (False risks)	**Strict Python-Only (Risk: None)**	**Strict Python-Only (Risk: None)**

3. Hard Acceptance Gates Audit Matrix

All statistical acceptance gates established prior to model deployment have been rigorously evaluated against the empirical data:

Gate ID	Target Metric & Criterion	Legacy v1 Result	Upgraded v2 Result	Audit Verdict
Gate 1A	OOD Pooled Brier Score ≤ 0.25	0.2413 (Unstable tails)	**0.2221 (Well-calibrated)**	**PASSED**
Gate 1B	Probability Tail Bounds (No 0.000/1.000)	0.0000 ≤ p ≤ 1.0000 (Pinned)	**0.1669 ≤ p ≤ 0.7779 (Smooth)**	**PASSED**
Gate 2A	Engine A 46-Repo LOPO ROC-AUC ≥ 0.65	N/A (Engine did not exist)	**0.7113 (Δ +0.0613 above gate)**	**PASSED**
Gate 2B	Engine A 46-Repo LOPO Lift ≥ +0.08	N/A (Engine did not exist)	**+0.1990 (Δ +0.1190 above gate)**	**PASSED**
Gate 3	Engine A Zero-Shot OOD ROC-AUC ≥ 0.65	N/A (Engine did not exist)	**0.7522 (Retains 98.0% of Engine B)**	**PASSED**

4. Expanded 4-Repository Zero-Shot Out-of-Distribution (OOD) Benchmark

To eliminate sample-size bias from the initial 2-repo OOD set, two diverse domain holdouts (**Pillow**: Imaging with native C-extensions, and **FastAPI**: Modern asynchronous web framework) were reserved. The zero-shot evaluation across all 1,680 unseen snapshots is shown below:

Repository	Domain	Snapshots	Prev %	Eng B PR-AUC	Eng B ROC	Eng B Lift	Eng A PR-AUC	Eng A ROC	Eng A Lift
django	Web ORM	804	78.1%	0.8976	0.7187	+0.1165	0.8867	0.6726	+0.1056
rich	Terminal UI	424	74.3%	0.9002	0.7527	+0.1572	0.8771	0.7268	+0.1342
pillow	Imaging C-Ext	165	41.8%	0.6459	0.7117	+0.2277	0.6040	0.6932	+0.1858
fastapi	Async Web API	287	20.9%	0.6273	0.8290	+0.4182	0.5758	0.8072	+0.3667
POOLED OOD (1,680)	Pooled Zero-Shot Pool	1680	63.8%	**0.8350**	**0.7580**	**+0.1969**	**0.8217**	**0.7522**	**+0.1836**

OOD GENERALIZATION VALIDATION

Key Takeaways from the 4-Repo OOD Expansion:

• **Engine A Zero-Shot Parity:** Engine A achieves **0.7522 ROC-AUC** and **+0.1836 Lift** on completely unseen repositories without using any Git churn history, retaining **98.0%** of Engine B's ROC-AUC (0.7580).

• **FastAPI Low-Prevalence Resilience:** On FastAPI (prevalence 20.91%), Engine B achieved **0.8290 ROC-AUC (+0.4182 Lift)** and Engine A achieved **0.8072 ROC-AUC (+0.3667 Lift)**, demonstrating exceptional discrimination on low-defect repositories.

5. 46-Repository LOPO Cross-Validation Benchmark & Lift Analysis

Leave-One-Project-Out (LOPO) cross-validation was executed across all 46 in-pool training codebases (21,490 snapshots). Macro-averaged results confirm that model lift remains positive across 100% of tested repositories:

Metric Dimension	Engine B (Full 73 Features)	Engine A (Static AST 34 Features)	Delta (AST Retention)
Mean Prevalence (Baseline)	43.27%	43.27%	0.0%
Mean PR-AUC	**0.6642**	**0.6318**	-0.0324 (95.1%)
Mean ROC-AUC	**0.7400**	**0.7113**	-0.0288 (96.1%)
Mean Empirical Lift	**+0.2314**	**+0.1990**	-0.0324
Mean Brier Score	**0.1982**	**0.2036**	+0.0054
Top 20% Inspection Precision	**70.07%**	**67.97%**	-2.10%
Top 20% Inspection Recall	**33.64%**	**32.50%**	-1.14%

Representative LOPO Sample Repositories (Spread of Domains & Prevalences)

Repository	Snapshots	Prev %	Eng B PR-AUC	Eng B ROC	Eng B Lift	Eng A PR-AUC	Eng A ROC	Eng A Lift
aiohttp	362	59.7%	0.7765	0.7047	+0.1798	0.7347	0.6725	+0.1380
black	614	59.1%	0.8567	0.7712	+0.2655	0.8454	0.7552	+0.2542
celery	660	41.8%	0.6573	0.7508	+0.2391	0.6006	0.7173	+0.1824
click	240	57.1%	0.8029	0.7403	+0.2321	0.7608	0.7093	+0.1899
httpx	731	27.8%	0.5364	0.7501	+0.2587	0.5451	0.7754	+0.2674
lightgbm	221	62.9%	0.8813	0.8183	+0.2523	0.8749	0.8030	+0.2459
mypy	843	59.5%	0.8660	0.8046	+0.2705	0.8561	0.7946	+0.2606
pytest	586	43.2%	0.6876	0.7716	+0.2558	0.6539	0.7424	+0.2222
scikit-learn	536	60.5%	0.8045	0.7416	+0.2001	0.7614	0.6915	+0.1570
tornado	823	37.1%	0.5839	0.7125	+0.2133	0.5377	0.6762	+0.1671

6. Production Integration, Strict Language Isolation & Latency

To ensure scientific rigor in production, strict language isolation and architecture guardrails were established:

PRODUCTION INTEGRATION CONTRACTS

1. Strict Python-Only Defect Risk Inference:
- **Python (.py):** Evaluates Dual-Engine ML model, producing calibrated defect probabilities and risk tiers.
- **Polyglot Source (.java, .ts, .js, .go, .rs, .c, etc.):** Returns factual static code telemetry (LOC, complexity) with risk: null.
- **Binary & Media (.jpg, .png, .pdf, .zip):** Returns zero counts with language: 'binary' and risk: null, eliminating false priors.

2. Live Inference Performance:
- **Mean Latency:** 75.98 ms across interactive IDE keystroke debouncing.
- **P90 Latency:** 86.12 ms (well within the sub-100ms interactive budget).

3. Architecture FFI Guardrails: Imports of cffi or ctypes automatically trigger an advisory confidence warning.

7. Final Audit Verdict & Reproducibility Certification

AUDIT CONCLUSION: FULL DEPLOYMENT APPROVAL

CERTIFICATION STATEMENT: The SmellPredict v2 Dual-Engine architecture completely resolves all 4 forensic audit findings: tail probability pinning has been eliminated via Platt Sigmoidal scaling; cold-start degradation has been eliminated by deploying a zero-git Engine A retaining 96.1% of ROC-AUC; prevalence inflation artifacts are transparently corrected via Lift metrics; and non-Python files are strictly protected against invalid risk inference.

Status: All 119 automated test cases pass (100%). Model serialized and deployed to production.